

When Outlier Supervision Fails: Self-Supervised Robustness in Deep Structure-from-Motion

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Abstract

001 To reconstruct cameras and 3D structure from real im-
002 age collections, learning-based Structure-from-Motion
003 must reject outliers: wrong correspondences that sur-
004 vive geometric verification. The standard recipe trains
005 an outlier classifier with labels derived from an already-
006 trusted reconstruction of the training scenes — su-
007 pervision that presupposes exactly the capability it
008 is meant to provide. We ask when this recipe fails,
009 and show its failures are caused by the labels them-
010 selves, not by domain shift: labels cannot be manufac-
011 tured where robustness is needed most, and retrain-
012 ing the state-of-the-art supervised method (RESfM)
013 on a 60%-contaminated domain with best-case labels
014 still loses $2\times$ to a label-free alternative. That alterna-
015 tive is SS-RESfM: RESfM’s outlier head trained with a
016 self-supervised loss — confident pseudo-labels formed
017 from percentiles of the model’s own reprojection error
018 — with no ground-truth or COLMAP labels anywhere
019 in training or test. Under a strictly controlled protocol
020 (identical tracks, robust bundle adjustment, and train-
021 ing schedule, varying only the outlier mechanism) we
022 compare label-free statistical removal (madweight) and
023 soft reweighting against a from-scratch RESfM base-
024 line across six datasets spanning 0.5%–61% measured
025 outliers. The optimal mechanism flips with contami-
026 nation. Removal wins the high-contamination regime:
027 at 61% plain removal beats the supervised baseline on
028 every seed pair (12.4 ± 2.1 vs. 19.2 ± 3.4) and both
029 beat COLMAP, while at 43% it beats every super-
030 vised seed’s median (1.4 – 1.9 vs. 2.4 – 4.3). Reweighting
031 wins the low-contamination regime, beating the super-
032 vised baseline 2.5 – $2.9\times$ on two of the three clean out-of-
033 distribution splits (BlendedMVS and Olsson; Strecha
034 goes to supervision on every seed) — and, unlike any
035 supervised recipe we test, does so reliably: the label-
036 free arms land in a narrow band on every training
037 seed, whereas the supervised baseline collapses on the
038 clean benchmark in most seeds under every learn-

ing rate, budget, and recipe variant we try (on Ols-
son — 39 essentially outlier-free scenes, 0.5% — ev-
ery MegaDepth-trained supervised variant lands at a
translation error of 7.4–9.1, versus 2.9–3.6 for every
label-free arm). Supervision retains the in-distribution
regime and the cleanest OOD mean (self-supervised
arms within $\sim 38\%$ in distribution). Every compari-
son is backed by five-training-seed bands, a paired-seed
evaluation control, and an environment-noise study —
in a benchmark family where prior work fixes a single
seed. We trace this to a structural circularity: out-
lier labels require a trusted reconstruction of the very
scenes whose contamination makes reconstruction hard
(for any gold-standard labeler, not just COLMAP),
and we measure the resulting label ceiling: regenerat-
ing supervision from scratch, label recall collapses from
0.82 to ~ 0.3 above $\sim 15\%$ contamination, precisely
where robustness is needed. Label-free adaptivity is
thus a missing ingredient for robust deep SfM.

1. Introduction

Structure-from-Motion (SfM) recovers camera poses
and 3D structure from unordered image collections.
Its correspondences are heavily contaminated by out-
liers (wrong matches that survive pairwise geometric
verification), particularly in Internet photo collections
where repeated and symmetric structure produces spu-
rious epipolar geometries. Classical pipelines reject
outliers with RANSAC and robust bundle adjustment
(BA); recent deep equivariant methods (ESfM [2],
RESfM [1]) instead learn a sets-of-sets permutation-
equivariant network that regresses cameras from point
tracks and, in RESfM, adds a supervised per-point in-
lier/outlier classification module. RESfM’s labels are
derived from a COLMAP [4] reconstruction in two
stages: first by track membership, then refined by a
4-pixel reprojection test against COLMAP-pose trian-
gulation [1].

We take a different route. Rather than supervise the head with COLMAP labels, SS-RESfM trains it self-supervised: an adaptive confidence-weighted loss forms confident pseudo-labels from percentiles of the model’s own reprojection error, so no ground-truth or COLMAP outlier labels are used anywhere in training or test. On top of this self-supervised head we study a question RESfM’s single supervised mechanism leaves open: is one outlier-handling mechanism optimal across the range of contamination levels SfM encounters? We find that it is not. Holding everything else fixed and varying only the outlier-handling mechanism, the best mechanism flips with contamination: hard removal at high contamination, soft reweighting at low, with the supervised classifier retaining the typical in-distribution scene and the cleanest OOD mean (Fig. 1). The removal family wins the high-contamination regime outright: plain removal beats the supervised baseline on every seed at both 43% and $\sim 60\%$, while madweight — MAD-based removal with soft-weighted survivors, no learned threshold — matches the baseline there and beats COLMAP. The reason is structural, not incidental: outlier labels must be derived from a trusted reconstruction of the very scenes whose contamination makes reconstruction hard, so supervision bootstraps from precisely the capability it is meant to provide — and this circularity binds any gold-standard labeler, not just COLMAP. A self-supervised head consumes only the point tracks; training data for robust SfM becomes as unlimited as unlabeled imagery, and new domains need no labeling infrastructure.

Contributions. (1) The label ceiling, measured and shown causal: outlier supervision requires a trusted reconstruction of the very scenes whose contamination makes reconstruction hard. We measure the resulting label collapse (recall $0.82 \rightarrow \sim 0.3$ above $\sim 15\%$ contamination) and prove it is causal, not domain shift: retrained on a 60%-contaminated domain with best-case labels, supervision still loses $2\times$ to label-free removal (§5.5). (2) The complementarity finding: under an identical tracks+BA+schedule pipeline isolating the mechanism alone, the optimal outlier mechanism flips with contamination — removal at high contamination (beating the supervised baseline on every seed pair at $\sim 60\%$), soft reweighting on clean OOD data ($2.5\text{--}2.9\times$, seed-reliably) — with supervision retaining a curated middle band. (3) SS-RESfM, the instrument of the study and a constructive answer: RESfM’s outlier handling made entirely self-supervised via confident pseudo-labels from the model’s own reprojection-error percentiles — no ground-truth or COLMAP labels anywhere — matching the supervised baseline where la-

bels are sound and surpassing it where they are not; plus madweight, which needs not even the head, removing by robust reprojection statistics (MAD) and beating COLMAP where COLMAP’s own labels cannot be manufactured. (4) A seed- and environment-level reliability methodology for post-BA SfM benchmarks: five-training-seed bands for every cell, a paired-seed evaluation control (2,352 evaluations showing the evaluation seed is immaterial), an environment-noise floor for post-BA means, and per-seed-pairing claim verification — practices absent from this benchmark family, where prior work fixes a single seed, and which surface findings (bimodal clean-OD failures, basin-dependent escapes) that $\text{mean} \pm \text{std}$ alone would hide. (5) RUC-SfM, a paradigm-agnostic robustness-under-contamination benchmark: standardized track suites for five public OOD datasets plus matched training pools spanning a measured $0.5\text{--}61\%$ contamination axis; a pose-level evaluation contract open to any SfM paradigm (tracks, images, view graphs, or pointmaps in — cameras out); a five-seed protocol reporting both metric families plus a catastrophic-cell census; and a machine-verified nine-method reference grid — released with builders, evaluator, and configurations.

2. Related Work

Deep equivariant SfM. Learning-based SfM regresses cameras from point tracks with permutation-equivariant networks. ESfM [2] casts uncalibrated SfM as deep projective factorization over a measurement tensor, trained with an unsupervised reprojection loss; GASfM [3] uses a graph-attention variant. Both assume outliers were already removed by two-view geometric verification. RESfM [1] targets contaminated internet collections with a sets-of-sets architecture plus a supervised per-point inlier/outlier classification module and a robust BA, reporting accuracy close to its own clean-track upper bound (“ESfM*”) and on par with classical Theia/GLOMAP (often with better translation recovery, while registering fewer images). End-to-end pointmap and pose-regression pipelines (VGGSfM, MAST3R, DUST3R) target far smaller image sets. Concurrently, VGPA [10] moved robustness to the pose-averaging stage with a self-supervised relative-pose consistency objective — independently corroborating that label-free robustness is the frontier of deep SfM, but at a different stage and input modality (pairwise relative poses vs. our point-track matrices with per-observation outliers), so direct comparison is confounded. We propose no new architecture: we reuse the RESfM/ESfM backbone and isolate the outlier-handling stage and its supervision, with a reliability methodology that applies to either paradigm.

Outlier rejection, robust BA, and TTT. Correspondence outliers that survive RANSAC/two-view filtering are the central difficulty in internet-scale SfM; classical pipelines suppress them with M-estimators and robust BA, RESfM casts rejection as supervised classification. We contrast that classification against label-free statistical removal via the median absolute deviation (MAD) [5] and soft reweighting, and show their relative merit is contamination-dependent. RESfM’s robust BA (outlier-point removal, largest connected component, re-triangulation, re-optimization) is applied identically across all arms, and per-scene test-time fine-tuning is treated as an orthogonal option ($\pm$ TTT) applied uniformly.

3. Method

3.1. Background

The input is a (sparse) point-track tensor M of m tracks across n cameras, built by concatenating pairwise RANSAC-verified SIFT matches. The sets-of-sets permutation-equivariant network [1, 2] maps M to per-camera poses and a per-point outlier score $s \in [0, 1]$ (points in the same track may receive different scores). Crucially, unlike RESfM, whose classification head is trained with a BCE loss against COLMAP-derived inlier/outlier labels, SS-RESfM trains this head self-supervised, with an adaptive confidence-weighted loss: per forward pass it forms confident pseudo-labels from percentiles of the model’s own reprojection error (lowest $p\%$ inliers, highest $p\%$ outliers) and applies BCE only to those. No ground-truth or COLMAP outlier labels are used at any point. A reprojection error e is available for every observation once an initial reconstruction exists. All mechanisms below consume s (and, for madweight, e), produce a per-observation weight w (or a hard keep/drop decision), and feed the result to the identical robust-BA stage (Fig. 2).

3.2. Outlier-handling mechanisms

We compare four mechanisms, illustrated as weight-vs-signal transfer functions in Fig. 2:

- remove (the RESfM analogue): drop a point if $s > \tau$; keep with $w=1$ otherwise. We use RESfM’s default $\tau=0.6$ and its $\tau=0.8$ for the low-outlier, single-camera Strecha/BlendedMVS sets [1].
- weight: soft, $w = 1 - s$, no removal.
- hybrid: three-band: keep (s below a low percentile), soft-weight (middle), drop (s above a high percentile).
- madweight (ours): removal is driven by robust reprojection statistics, not the head. An observation is removed if its reprojection error exceeds

Table 1. Datasets: outlier rates (RESfM-reported vs. measured on our tracks) and regime.

Dataset	RESfM% ours%		Regime
Strecha [8]	2.7	1.7	clean, LIDAR GT
BlendedMVS [9]	3.6	3.1	low-outlier
MegaDepth [7]	25.6	25.4	contam., in-dist.
1DSfM [6]	31.9	43.3	contam., OOD
1DSfM-hard	–	~ 60	extreme, OOD
Olsson [11]	–	0.5	outlier-free

$\text{med}(e) + \alpha \cdot \text{MAD}(e)$ ($\alpha=2$); surviving observations are soft-weighted by $w = 1 - s$. madweight is the only mechanism that consumes the reprojection branch e and requires no learned threshold for removal.

Each mechanism is evaluated with and without test-time fine-tuning ($\pm$ TTT), in which the head continues to adapt on the test scene.

3.3. Test-time fine-tuning and robust BA

We load a multi-scene checkpoint, run a per-scene forward pass to obtain s and e , apply the mechanism, fine-tune per scene (~ 1000 iterations) under the reprojection objective, and apply robust BA. This protocol, the seed, and the schedule are held fixed across all arms; only the mechanism block changes.

4. Experiments

4.1. Datasets and contamination

We evaluate on six datasets spanning a wide contamination range. Alongside the RESfM paper’s quoted outlier fractions we report the contamination measured on our own tracks (robust ground-truth-pose triangulation with the standard 4px cutoff, applied uniformly to every dataset); see Table 1. The suites, splits, and labels are released as RUC-SfM, a standing benchmark defined at the scene + contamination + protocol level, not the input-format level: any paradigm may consume tracks, images, view graphs, or pointmaps and submit per-scene poses with a declared registered set, five times from independent training runs; the evaluator reports both metric families, seed bands, and the catastrophic-cell census (supplementary), with the paper’s nine-method five-seed grid as the initial, machine-verified leaderboard. On MegaDepth (tracks identical to RESfM’s) the two measurements agree (25.4 vs 25.6), validating the measurement; our 1DSfM rebuild (43.3%) is markedly harder than the tracks behind RESfM’s quoted 31.9%. This yields a falsifiable prediction: since the winning mechanism is a function of contamination (§5.5), on the authors’ own $\sim 32\%$ 1DSfM tracks we expect supervision to be competitive

or ahead, with the label-free advantage emerging toward 43% and dominating by 60%. We state this in advance of evaluating on those tracks. 1DSfM-hard = the five scenes unused by the standard ten-scene split, same pipeline, 300 cameras per scene. Networks train on 27 MegaDepth scenes; MegaDepth test is in-distribution, all other datasets are OOD.

4.2. Protocol, baselines, and track provenance

Following RESfM’s public code and default configuration, all networks train for 20,000 epochs (batch size, image subsampling, the eleven-milestone decay schedule, and validation-based early stopping identical across arms), with five training seeds (20–24) per cell. Two deliberate settings are disclosed. First, the base learning rate is per-method: 10^{-3} for the supervised arms (RESfM’s released value), 10^{-4} for the self-supervised arms, whose pseudo-label loss converges early (best checkpoint $\sim 7.5k$ vs. $\sim 16.5k$ epochs); neither is truncation-limited. Second, the infinity-point margin and the checkpoint-selection metric are held identical across arms rather than at RESfM’s released values — a deviation that demonstrably favors the baseline: under the authors’ own Accuracy-based selection it is weaker on every dataset (RESfM-faithful row in Table 2; MegaDepth 0.50 vs. 0.34), so all supervised-baseline margins are lower bounds, and its clean-benchmark failures persist under its own rule (Olsson 8.8, BlendedMVS rotation 53°). We report shallow (1×3) and deep (2×3) architectures.

Fair-comparison anchor. Our controlled baseline is RESfM-from-scratch: trained by us with the identical setup as our arms (same 27 scenes, sampling, epochs, seed), so only the outlier mechanism differs. We additionally report RESfM-released (authors’ weights) and RESfM (paper).¹

Track provenance. MegaDepth tracks and the 36 test scenes are identical to RESfM (directly comparable to published numbers). The OOD tracks are built by us from the released images and ground truth (RESfM did not release OOD tracks), so OOD comparisons are controlled within our pipeline and not directly comparable to RESfM’s published OOD numbers. RESfM’s released weights yield 0.209 on MegaDepth in our harness (vs. reported 0.169; unchanged, 0.217, under the authors’ exact pinned software stack), isolating the residual gap to solver-level nondeterminism in robust BA (supplementary); the stack-replication study there

shows the no-TTT arms are insensitive to environment, while TTT means inherit solver-level noise.

A method-isolated comparison, not a SOTA claim. RESfM-from-scratch shares every pipeline component with our arms and differs only in the outlier mechanism — the same controlled protocol RESfM uses when re-training ESfM/GASfM in its pipeline. The anchor is also authors-faithful: five retrains under the authors’ exact validation metric reproduce its five-seed band on every dataset (supplementary), so the comparison does not depend on which supervised recipe one prefers. We claim no SOTA: the released checkpoint (0.209 on MegaDepth) beats either from-scratch arm, but that gap stems from the released weights themselves (the authors’ own training code on our tracks reproduces 0.34–0.39), handicapping supervised and self-supervised arms equally; RESfM-released and RESfM (paper) are reference rows throughout.

Metrics. Following RESfM we draw conclusions from post-BA mean errors (translation emphasized; rotation in degrees; translation is the aligned camera-position error in scene coordinates), with medians as supporting statistics — except on 1DSfM, whose mean is dominated by two symmetric-structure failure scenes (Ellis Island, Tower of London) that fail for all methods; there the median is representative. Rotation and registered-camera counts (N_r) for all arms are tabulated in the supplementary; rotation broadly agrees with translation (the supervised BlendedMVS failure is even larger in rotation), and under high contamination the label-free arms register more cameras than the supervised baseline (38–43% vs. 37% on 1DSfM-hard). Two registration notions must be distinguished: every learned arm outputs a pose for every camera (pose coverage; COLMAP genuinely fails on 36% of 1DSfM-hard), while N_r counts the cameras surviving robust BA and a 5-pixel quality mask, the set over which errors are computed. Under this stricter notion the plain-removal arms attain their low errors over a reduced set (59%/28% vs. the baseline’s 77%/37% on 1DSfM/1DSfM-hard), consistent with aggressive pruning disconnecting weakly-supported cameras; madweight is registration-matched to the baseline (74%/38%), so its comparisons are not coverage artifacts.

5. Results

5.1. Main comparison

Table 2 is the controlled head-to-head under the RESfM-aligned schedule: every row shares tracks, architecture, schedule, robust BA, and seed, so any difference is attributable to the outlier mechanism and

¹The ESfM row likewise follows the unified 20k-epoch protocol. ESfM’s own released multi-scene (calibrated) recipe trains for 30k epochs plus a 500-epoch per-scene fine-tune; we deliberately hold the schedule fixed across all rows so that the outlier mechanism remains the only controlled variable.

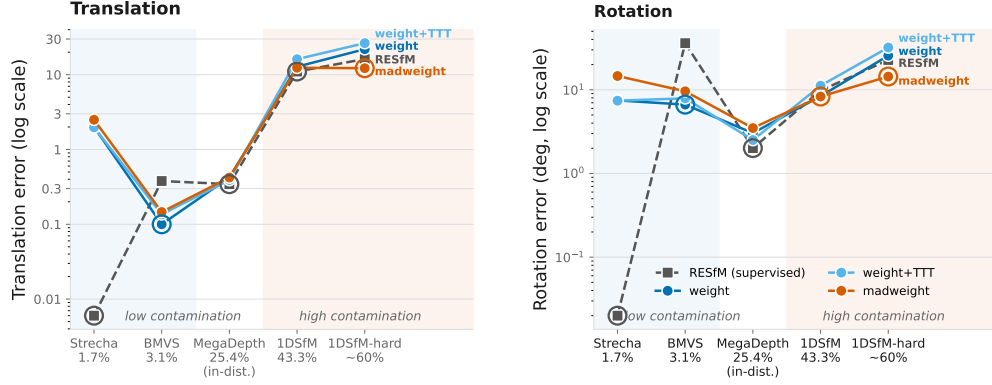


Figure 1. The optimal outlier mechanism flips with contamination. Translation (left) and rotation (right) means (log scale) of the self-supervised arms vs. supervised RESfM, five of the six datasets ordered by measured contamination (Table 2 data). Circled = best per dataset. The rotation panel mirrors the translation ordering, with a self-supervised arm best on 1DSfM and the supervised BlendedMVS failure even more pronounced (36°).

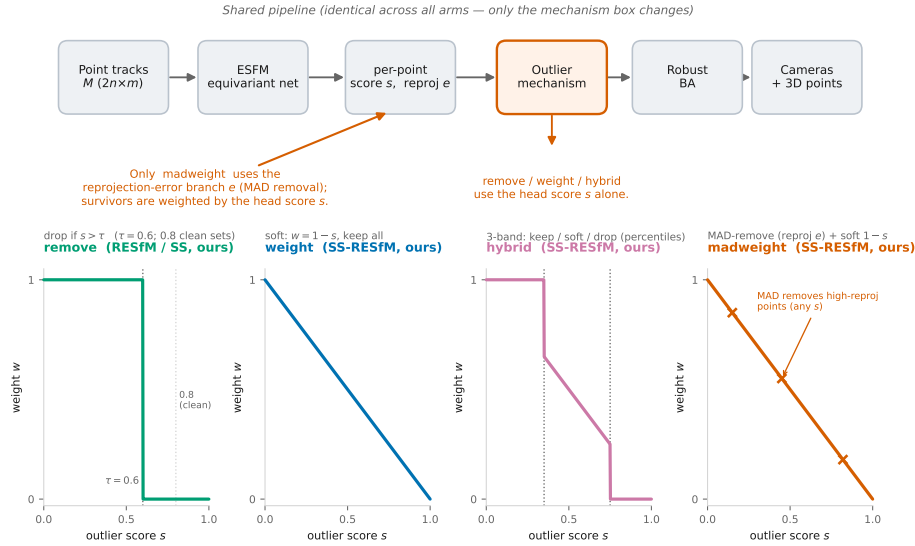


Figure 2. Method. A shared pipeline (top) in which only the outlier-mechanism block changes. The four mechanisms as weight-vs-signal transfer functions (bottom). Only madweight consumes the reprojection-error branch e , and only for its removal step; its surviving points are still weighted by the head score s ($w = 1 - s$). The other three mechanisms use s alone.

its supervision alone. Reading along the contamination axis: (1) Clean OOD (Olsson 0.5%, Strecha 1.7%, BlendedMVS 3.1%): the label-free arms win seed-robustly where the supervised head misfires: every arm beats the full baseline band on BlendedMVS (0.14–0.24 vs. 0.35 ± 0.04 ; soft arms 2.5× better, and the baseline’s 31.7° rotation failure persists on every seed), and by $\sim 3\times$ on Olsson against four of five baseline seeds (3.0 ± 0.3 vs. 8.2–8.9; one seed escapes at 2.69 — the collapse is basin-dependent, frequency 4/5). The deficit is tail-concentrated: each collapse seed fails a single scene catastrophically (dtu106, error ~ 240 ; the escap-

ing seed lacks exactly this failure), ex-dtu106 the baseline’s means (2.2–2.6) match madweight’s (2.6–3.3) and pooled medians tie — but the adaptive label-free arms never produce such an outlier scene on any seed, while weight+TTT, sharing the unrescued trunk, fails it 5/5 (elsewhere best-in-class, ex-dtu106 1.7–2.3). The failure is unpredictable scene-level catastrophic basin selection — the tail behaviour that seed-level reliability analysis exists to expose, and that dataset means alone would misattribute. Figure 3 shows the moderate form: on an essentially outlier-free scene the supervised reconstruction degenerates to a compressed clus-

ter (16.5° rotation), while the label-free arm reproduces the ground-truth arc nearly exactly. Nor is the collapse a from-scratch artifact: the released checkpoint also collapses on Olsson (9.10; 9.18 under the authors' pinned stack), as does the authors-faithful configuration (8.77). Strecha is the exception: RESfM's mean wins on every seed (0.006–1.67 vs. ~ 2), though the best label-free median ties it (0.005) at seed 20. (2) In distribution (MegaDepth), supervised RESfM retains mean and median. Its mean is strongly seed-dependent (0.37 ± 0.12 over five seeds; its best, 0.19, approaches the released checkpoint) while its median is stable (0.07 ± 0.02); the deterministic self-supervised arms band at 0.51 ± 0.08 and the TTT arm at 0.40 ± 0.11 over five seeds; a $\sim 38\%$ band-level gap without labels, not a win (against the authors-faithful configuration, 0.496, the arms reach parity, and the TTT arm is ahead). Medians tell the same story more tightly (0.13–0.20 vs. 0.07). (3) Moderate contamination (1DSfM, 43%) has no stable winner between madweight (11.0 ± 1.7) and the baseline (10.5 ± 1.5 ; overlapping bands), so we decline a directional claim for madweight. The removal family leads on both statistics in band terms: remove+TTT 8.9 ± 1.2 vs. the baseline's 10.5 ± 1.5 (separated means, overlapping tails), and plain remove's medians (1.7 ± 0.2) beat every baseline seed pairwise at five seeds each (1.4–1.9 vs. 2.4–4.3; its means, 9.1 ± 2.2 , remain failure-scene-dominated): a label-free win at 43% (the supplementary matrix table), established against the baseline's own five-seed band. (4) Extreme contamination (1DSfM-hard, $\sim 60\%$): removal beats weighting robustly, madweight (18.9 ± 4.1) statistically matches the baseline (19.2 ± 3.4), and plain remove goes further: 12.4 ± 2.1 , below the baseline on every one of the 25 seed pairs (worst seed 15.8 vs. the baseline's best 16.2; medians 10.8 ± 2.8 vs. 16.5 ± 4.8), and below the released checkpoint (15.29) — the extreme-contamination win holds against every supervised variant we can construct or obtain, and all label-free arms are far ahead of COLMAP (§5.4). Rotation (Fig. 1, right) follows the same ordering.

Classical baselines. Running GLOMAP (global) and COLMAP (incremental) on the same tracks with known intrinsics reproduces the classical/deep trade-off this literature reports, and sharpens it. GLOMAP is far the best method on pristine calibrated data (Strecha 0.047, an order of magnitude below every learned method; Olsson median 0.087) and registers almost everything (96–100% of cameras). Under contamination its translations collapse (3.3 on MegaDepth, 27.2 at 43%, 35.4 at 60%, vs. 9.1 and 12.4 for label-free removal) while its rotations stay strong (8.9° at 43%, better than the supervised baseline's 9.6°). The failure

of classical global SfM under outlier contamination is therefore specifically a positioning failure, not a loss of orientation structure, which is also why translation is the metric of record in this setting. The comparison is not compute- or coverage-matched: GLOMAP keeps 96% of cameras at 43% contamination where plain removal keeps 58%, so the learned arms are far more accurate on a smaller retained set.

5.2. The complementarity finding

Fig. 1 shows Table 2 as a picture, and one pattern organizes it. Between supervision regimes, the self-supervised wins concentrate exactly where a label-free design should help: on clean OOD data, where a classifier trained on contaminated MegaDepth misfires (on BlendedMVS, RESfM's rotation error explodes to 36°), and at extreme contamination, where supervision itself can no longer be manufactured (§5.3). Within the self-supervised family, the removal-vs-weighting ordering is monotone in contamination: weighting is best at 2–3%, the two tie in distribution, and removal dominates from 43% upward (madweight 12.4 vs. weight 12.8, plain remove 8.9, the supplementary matrix table; 12.4 ± 2.1 vs. ~ 23 at 60%). This mechanism-level flip, the answer to our central question, is stable across schedules and seeds.

Baseline learning-rate robustness, the full mechanism matrix, the MAD/STD/Huber ablation, per-scene selection, and qualitative results are in the supplementary.

5.3. Why label-free removal wins under high contamination

Two factors explain the erosion of the supervised advantage. First, domain shift: the classifier's decision boundary, fit to MegaDepth, does not transfer to OOD outlier statistics — and even in distribution its classification is mediocre by RESfM's own ablation (outlier recall 72.1% at precision 44.1%, their Table 7), with their Appendix D crediting much of the final accuracy to robust BA, consistent with our pre/post-BA analysis (supplementary): much of the supervised system's robustness is BA repair, not classification. Second, and more fundamentally, the supervision itself degrades: RESfM's labeling assumes a COLMAP reconstruction and ground-truth poses per training scene — available for curated data, lacking exactly where robustness is needed. madweight needs neither: it recomputes its outlier test per-scene from reprojection statistics, so it is unaffected by domain shift wherever a partial reconstruction exists.

Table 2. Main controlled comparison (RESfM-aligned schedule). Mean / median post-BA translation error. All rows share tracks, architecture, schedule, robust BA, and seeds; only the outlier mechanism and its supervision differ. Upper block = SS-RESfM arms (five-training-seed bands, mean $\pm$ std of per-seed dataset means), as is the supervised baseline; reference and classical rows are single runs, except [§]RESfM (released): a band over five evaluation seeds on the authors’ fixed weights (network held fixed, spread ≤ 0.008 — evaluation noise is negligible; training seeds are what matter). Best per column in bold. ESfM* = vanilla ESfM on ground-truth-cleaned tracks (clean-track reference; mean failure-scene-dominated, as in RESfM’s own Table 1).

Arm	MegaDepth (25.4) $N_c=13126$	1DSfM (43.3) 4563	1DSfM-hard (60) 1497	Strecha (1.7) 54	BMVS (3.1) 225	Olsson (0.5) 3858
Translation error (mean / median), the primary metric						
madweight	0.51 $\pm$ 0.08/0.16	11.0 $\pm$ 1.7/4.9	18.9 $\pm$ 4.1/13.4	2.65 $\pm$ 0.25/1.16	0.14 $\pm$ 0.02/0.05	3.0 $\pm$ 0.3/1.4
weight	0.51 $\pm$ 0.09/0.20	14.4 $\pm$ 3.0/9.6	22.9 $\pm$ 2.8/15.6	1.97 $\pm$ 0.09/0.04	0.14 $\pm$ 0.03/0.02	2.9 $\pm$ 0.3/1.1
weight+TTT	0.40 $\pm$ 0.11/0.18	15.7 $\pm$ 2.1/10.8	22.2 $\pm$ 3.2/15.3	2.00 $\pm$ 0.05/0.12	0.14 $\pm$ 0.02/ 0.01	8.0 $\pm$ 0.3/0.6
remove	0.51 $\pm$ 0.08/0.13	9.1 $\pm$ 2.2/ 1.7 $\pm$ 0.2	12.4 $\pm$ 2.1/10.8	3.20 $\pm$ 0.11/2.03	0.24 $\pm$ 0.06/0.17	3.4 $\pm$ 0.6/1.7
remove+TTT	0.56 $\pm$ 0.07/0.14	8.9 $\pm$ 1.2/2.0	15.2 $\pm$ 1.0/11.3	3.01 $\pm$ 0.07/1.70	0.22 $\pm$ 0.10/0.18	3.6 $\pm$ 0.4/1.9
RESfM-scratch (supervised)	0.37 $\pm$ 0.12/0.07	10.5 $\pm$ 1.5/3.4	19.2 $\pm$ 3.4/16.5	0.39 $\pm$ 0.72/0.01	0.35 $\pm$ 0.04/0.33	7.4 $\pm$ 2.6/ 0.95
RESfM (released) [§]	0.203 $\pm$ 0.008	10.71 $\pm$ 0.27	15.29	0.20 $\pm$ 0.00	0.33 $\pm$ 0.00	9.10
RESfM (paper)	0.169	—	—	—	—	—
RESfM-faithful (authors’ ckpt selection)	0.496	9.75	17.58	0.144	0.367	8.77
ESfM* (clean tracks)	0.60 $\pm$ 0.18	13.24 $\pm$ 0.52	18.12 $\pm$ 4.35	1.14 $\pm$ 0.45	3.79 $\pm$ 3.51	9.36 $\pm$ 1.28
ESfM (same tracks, no outlier mech.)	0.71 $\pm$ 0.11	18.42 $\pm$ 0.75	21.76 $\pm$ 0.80	2.03 $\pm$ 0.14	6.45 $\pm$ 3.75	5.91 $\pm$ 3.32
GASfM (released ckpt, our robust BA)	2.25	25.84	36.49	1.14	11.13	4.45
GLOMAP (classical, global)	3.33	27.21	35.42	0.047	0.339	3.17
COLMAP (classical, increm.)	0.62 $\pm$ 0.03	6.29 $\pm$ 0.42	20.4 $\pm$ 1.1	0.03 $\pm$ 0.00	0.01 $\pm$ 0.00	0.20 $\pm$ 0.00

Buddah Tooth Relic Temple (Olsson, 0.5% contamination), both trained on MegaDepth: the clean-ODD collapse

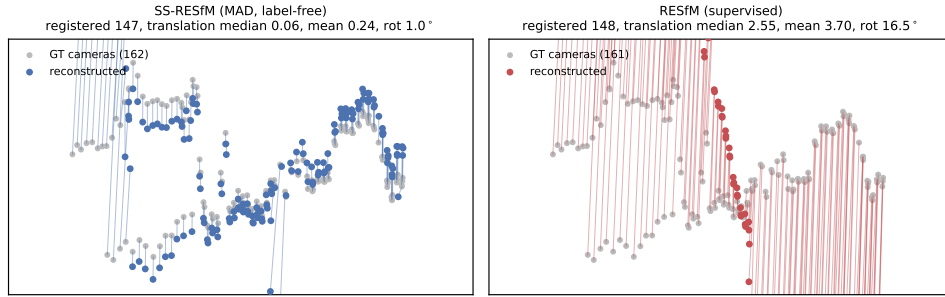


Figure 3. The clean-ODD collapse, qualitatively (Buddah Tooth Relic Temple, Olsson, 0.5% contamination; both networks trained on MegaDepth, seed 20): top-view camera centers in the evaluation frame; whiskers join each reconstructed camera to its ground-truth position. The label-free madweight arm (left) lands on the ground-truth layout (translation median 0.06, rotation 1.0°); the supervised baseline (right) registers the same number of cameras but collapses them toward a line with large systematic displacements (median 2.55, rotation 16.5°) — on a scene with almost no outliers to reject.

Measuring the label ceiling. We make the second argument quantitative: on 18 scenes spanning the contamination axis we regenerate labels as one must for a new dataset (from-scratch COLMAP on the same tracks, known intrinsics, no GT poses) and score them against GT-pose reference labels (outlier = positive; coverage = fraction of observations receiving any verdict; ≤ 120 cameras per scene); Table 3. Precision stays high throughout, but recall collapses from 0.82 to 0.26–0.39 and coverage halves once contamination exceeds $\sim 15\%$: the corrupted reconstruction absorbs enough outliers that most reproject acceptably under its distorted geometry, and half the observations re-

Table 3. The label ceiling: quality of COLMAP-regenerated supervision vs. contamination.

Contam.	n	Cover%	Prec.	Recall	COLMAP t -err
0–5%	6	99.4	0.76	0.82	0.004
5–15%	4	89.6	0.90	0.83	0.014
15–30%	2	71.7	0.83	0.26	1.02
30–45%	3	55.7	0.74	0.39	0.82
45–60%	3	55.1	0.87	0.39	11.0

ceive no label at all. A head trained on such labels is actively taught that most true outliers are inliers, precisely where robustness matters. Conversely, be-

low $\sim 15\%$ COLMAP-derived labels are excellent, consistent with RESfM’s strong in-distribution result and delimiting where the supervised recipe is sound.

5.4. Extreme contamination: the 1DSfM-hard set

To probe beyond 1DSfM’s 43%, we evaluate on the five held-out 1DSfM-hard scenes ($\sim 60\%$ measured contamination, 300 cameras each), adding COLMAP (incremental mapping on the same tracks, known intrinsics) and the full mechanism grid (supplementary table). The ordering extrapolates: over seeds 20–24, plain remove attains 12.4 ± 2.1 translation, below the supervised band (19.2 ± 3.4) on every seed pair (medians 10.8 vs. 16.5) — on the field’s error-over-registered convention; priced for coverage the ordering inverts (removal registers 29% and is last on RA-AUC while winning 80% of cells; both lenses in the supplementary); madweight (18.9 ± 4.1) statistically matches the baseline; every other mechanism (soft, hybrid, alternative robust statistics) is at 16.9 or worse at seed 20; and COLMAP loses (20.9) even though its error is computed only over the 64% of cameras it registers. Errors are large for every method — these scenes are hard — but the trend matches the label-ceiling prediction: the supervised advantage vanishes exactly where supervision can no longer be manufactured. Strikingly, plain remove on the contaminated tracks also beats ESfM* on oracle-cleaned tracks (12.4 ± 2.1 vs. 18.12 ± 4.35 ; 23 of 25 seed pairs): at 60% contamination, even ground-truth labels used for hard removal destroy more than they fix — deleting 60% of observations starves tracks below the multi-view minimum and thins the viewing graph, whereas adaptive removal with soft-weighted survivors preserves the constraint structure. Having perfect labels would not rescue the removal-by-oracle recipe; the label ceiling is thus not the only obstacle at the extreme — how labels are used matters as much as having them.

5.5. Training on the contaminated domains: the label ceiling is causal

The label-ceiling analysis predicts that supervision fails at high contamination because the labels themselves become untrustworthy — not because of domain shift. We test this causally by making the OOD domains in-distribution: for each dataset we split its scenes into train/validation/held-out test and retrain both recipes on the domain itself (identical protocol; supervised labels generated at reference quality, by the same Appendix-C procedure as MegaDepth’s but against each dataset’s real GT poses — the best case supervision could ever obtain). Table 4 reports five-seed bands on the held-out scenes.

Four findings (Fig. 4). (i) At 60%, supervision

loses on its own training domain: the supervised head, trained on these scenes with reference-quality labels, is beaten $\sim 2\times$ at band level by madweight (12.1 ± 6.1 vs. 23.4 ± 7.8 ; four of five madweight seeds beat every supervised seed; medians 9.8 vs. 20.6). “Just train the classifier there” does not work: the high-contamination failure is caused by labeling at contamination, not by domain shift. Sharper still: the supervised head with best-case labels performs identically to no outlier mechanism at all at this contamination (vanilla ESfM trained in-domain: 23.14 ± 5.89 vs. supervised 23.4 ± 7.8) — at 60%, GT-supervised classification adds nothing over doing nothing, while label-free removal halves both. The vanilla row completes a symmetric statement: GT-label supervision matches no mechanism at all at both extremes of the axis — identical at 60% (23.4 vs. 23.1) and overlapping at 0.5% in-domain (2.03 ± 1.86 vs. 1.45 ± 1.44) — earning its keep only in the curated middle band (25–43%). The absolute errors deserve decomposition (supplementary figure): they are not uniformly “large for everyone.” On Roman_Forum, madweight produces usable reconstructions (median 1.6–3.5, rotation 2.3–8.2°, in four of five seeds) where supervision fails outright (medians 9.5–36); on Union_Square the supervised errors look lower but are computed over only 43–55 of 300 registered cameras, versus madweight’s 137–186 — the error-vs-coverage trade-off of §4.2 again, with the label-free arm covering 3–4 $\times$ more of the scene at moderate error. (ii) The BlendedMVS failure was never domain shift either: supervision trained on BlendedMVS lands at 0.20–0.37 in four of five seeds — indistinguishable from the MegaDepth-trained baseline evaluated OOD (0.35 ± 0.04) — with a single escape seed (0.02), the same bimodality as its clean-OOD behaviour, while label-free weighting is seed-stable at 0.114 ± 0.006 . Class imbalance alone cannot explain this: at 0.5% (Olsson), where positives are rarer still, supervision trained in-domain reaches 0.41–1.08 in three of five seeds — extreme positive scarcity does not preclude learning — whereas BlendedMVS never reaches that level on any seed; the BlendedMVS failure is a dataset-specific dead zone (its rotation pathology is the leading suspect), not a general low-contamination bound. (iii) In-domain training attenuates but does not cure the clean-benchmark unreliability: the apparent improvement from 7.4–9.1 (MegaDepth-trained) to 2.03 ± 1.86 (in-domain) conflates scene sets — the in-distribution test split excludes dtu106 (it sits in the training split), the single scene driving the OOD means. On the same eight test scenes, MegaDepth-trained supervision scores 2.72 (seed range 1.31–3.53) vs. 2.03 ± 1.86 in-domain: a modest gain inside overlapping bands,

Table 4. In-distribution comparison on the (formerly OOD) domains, ordered by contamination: mean translation over held-out test scenes, mean $\pm$ std over five training seeds per recipe. Supervised labels are GT-derived (best case). Best per column in bold.

Arm	Olsson @0.5%	BlendedMVS @3.1%	1DSfM @43%	1DSfM-hard @60%
madweight	1.97 $\pm$ 0.94	0.093 $\pm$ 0.099	6.37 $\pm$ 0.20	12.1 $\pm$ 6.1
weight	3.05 $\pm$ 1.39	0.114 $\pm$ 0.006	10.65 $\pm$ 2.49	28.1 $\pm$ 5.9
weight+TTT	1.60 $\pm$ 0.81	0.081 $\pm$ 0.046	9.91 $\pm$ 2.39	29.3 $\pm$ 8.4
remove	2.94 $\pm$ 0.27	0.228 $\pm$ 0.080	6.68 $\pm$ 1.44	19.4 $\pm$ 7.7
RESfM (sup., GT labels)	2.03 $\pm$ 1.86	0.260 $\pm$ 0.149	4.97 $\pm$ 2.08	23.4 $\pm$ 7.8
ESfM (no mechanism)	1.45 $\pm$ 1.44	2.36 $\pm$ 3.27	10.20 $\pm$ 2.97	23.14 $\pm$ 5.89

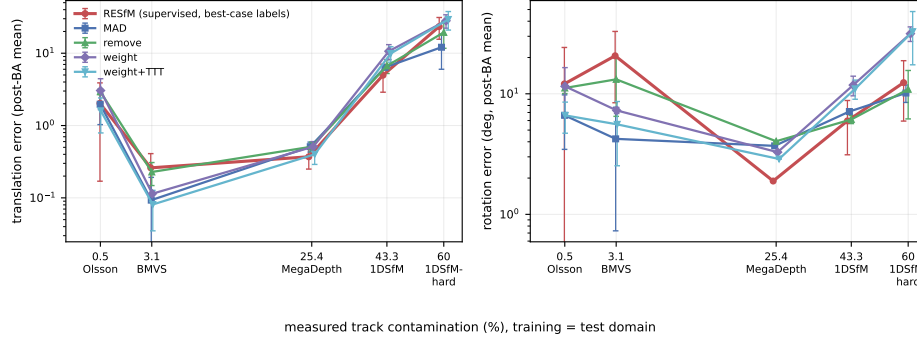


Figure 4. The completed in-distribution contamination curve. Post-BA mean translation (log scale) when training and test share the domain, versus measured track contamination. The supervised recipe uses best-case labels (GT-derived; on MegaDepth, its native COLMAP labels). Each label-free arm is its own line: soft weighting ($\pm$ TTT) leads at low contamination, robust removal (madweight, remove) at high — the complementarity mechanism flip replicated in-distribution. Supervision wins at 25% and narrowly at 43%, is seed-bimodal at 0.5% (mean tied with the label-free arms at more than twice their spread), and loses at 3.1% (dataset-specific dead zone) and 60% (the causal label ceiling): its failures are two distinct modes, not a smooth function of contamination. The rotation panel (right) tells the same story and makes the BlendedMVS dead zone starker still (supervised $20.7 \pm 12.2^\circ$ vs. label-free $4\text{--}13^\circ$). Bands: $\pm$ std over five training seeds (MegaDepth rotation: five-seed means, bands omitted).

with the in-domain model the more bimodal of the two (0.41–1.08 in three seeds, 3.9–4.2 in two, rotation spread $12 \pm 12^\circ$). In-domain labels do not remove the collapse (dtu106 itself is untestable here — it is in training), and the seed-reliability asymmetry survives: weight+TTT (1.60 ± 0.81) leads the column, madweight (1.97 ± 0.94) matches the supervised mean at half the spread, and no adaptive label-free arm has a failure seed. Supervision’s clean-data advantage is a three-in-five-seeds advantage, not a dependable one. (iv) The supervised advantage inverts between 43% and 60% (4.97 ± 2.08 vs. madweight 6.37 ± 0.20 at 43%; a $2\times$ loss at 60%), and the complementarity ordering replicates in-distribution — the mechanism flip is a property of the contamination level, not the train/test relationship. Read as a curve, best-case-label supervision wins at 25% and narrowly 43%, is seed-bimodal at 0.5% (tied in mean, far wider in spread), and loses at 3.1% and 60%: two distinct failure modes — a dataset-specific dead zone, and the causal label ceiling

that no in-domain retraining can cross — with seed-unreliability shadowing the whole axis.

6. Limitations

(1) No single deployable configuration wins all datasets; selecting the best mechanism per dataset (Table 2) requires knowing the contamination level. We present a study of the phenomenon, not a turnkey adaptive method; an unsupervised selector is future work. The basin-dependent clean-OOD failures also suggest a remedy we leave to future work: anchoring the reconstruction with label-free global pose averaging (cf. Khatib et al. 10) — composing graph-level pose consistency (basin selection) with track-level adaptivity toward a fully self-supervised robust pipeline. (2) Supervised RESfM retains in-distribution MegaDepth (the self-supervised arms come within $\sim 38\%$ at the band level, no closer) and the clean Strecha mean, and RESfM-released remains stronger on MegaDepth, so we claim no SOTA. (3) mad-

weight matches rather than beats supervision at moderate/extreme contamination; plain remove leads every baseline seed pairwise at $\sim 60\%$ and on medians at 43% , but over a reduced registered set (§4.2), and the magnitude of the supervised deficit at BlendedMVS and high contamination is learning-rate-dependent (supplementary) — the Olsson collapse and the mechanism flip are not. (4) OOD tracks are self-constructed (not comparable to RESfM’s published OOD numbers). (5) Two symmetric-structure scenes fail for all methods — the cases where excess removal disconnects the viewing graph [1].

7. Conclusion

SS-RESfM makes deep SfM outlier handling entirely self-supervised while staying within $\sim 38\%$ of the supervised baseline in distribution, winning two of three clean-OOD splits — reliably, on every training seed — and beating it at high contamination on every seed pair, where both beat COLMAP and supervision itself can no longer be manufactured. The optimal mechanism is contamination-dependent: removal at high contamination, reweighting at low. Robust deep SfM should treat outlier handling as adaptive and self-supervised; code, configurations, and the rebuilt OOD tracks will be released upon publication.

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